MOTION DETECTION BASED SECURITY SYSTEM

Using ESP32 and PIR Motion Sensor

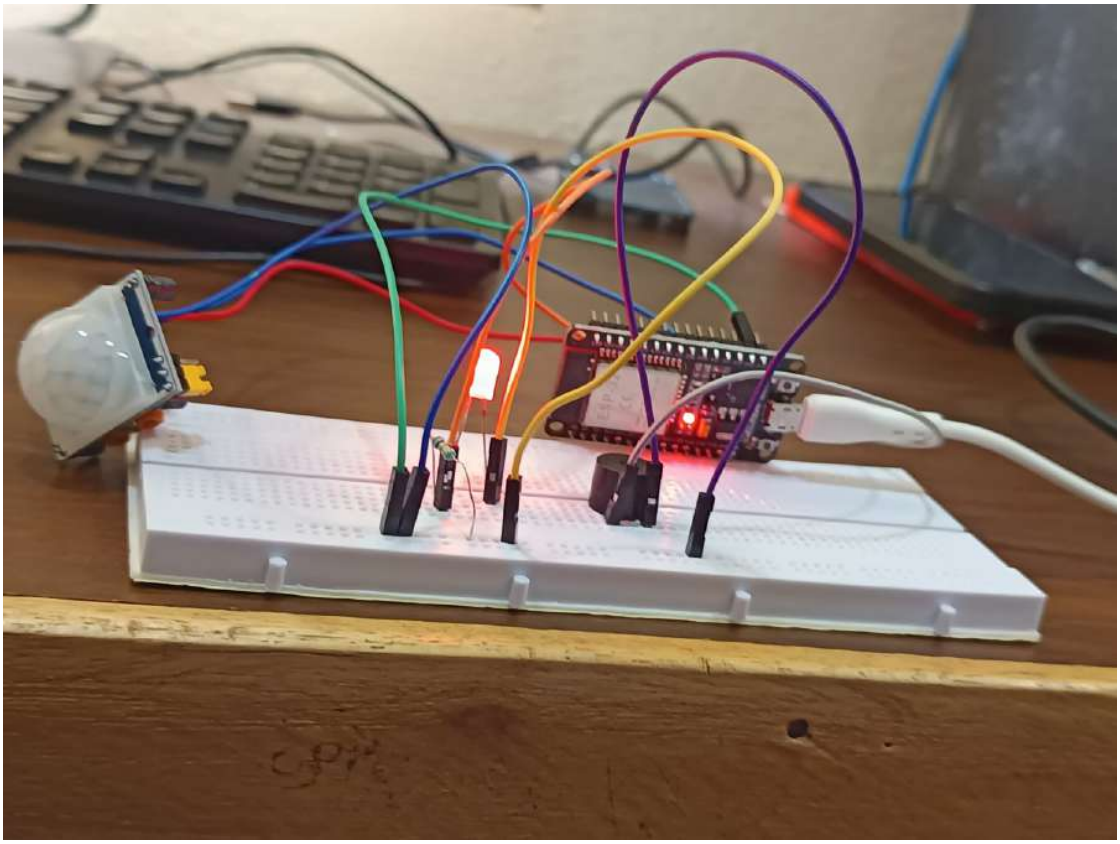


Figure 1: Hardware prototype of the ESP32-based motion detection system

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ABSTRACT

This project presents a simple motion detection-based security system developed using an ESP32 development board and a Passive Infrared (PIR) motion sensor. The PIR sensor monitors the surrounding area for changes in infrared radiation caused by the movement of a person or other warm object. Its digital output is read by the ESP32. When motion is detected, the sensor output becomes HIGH (logic 1), and the system reports “MOTION DETECTED!” through the Serial Monitor. When no motion is detected, the output is LOW (logic 0), and the system reports “NO MOTION”. An LED is also visible in the prototype as a visual indicator. The project demonstrates how a low-cost microcontroller and sensor can be used as the foundation of a basic security monitoring system.

1. INTRODUCTION

Security systems are used to monitor spaces and alert users when unexpected movement occurs. A motion detection system can help monitor rooms, laboratories, offices, storage areas, and other locations. In this project, an ESP32 microcontroller is interfaced with a PIR sensor. The ESP32 continuously checks the sensor's digital output and displays the detected condition on a computer through the Arduino IDE Serial Monitor.

2. OBJECTIVES

- To interface a PIR motion sensor with an ESP32 development board.
- To detect movement within the sensing range of the PIR sensor.
- To read and interpret the sensor's digital output as HIGH or LOW.
- To display motion status on the Serial Monitor.
- To demonstrate a basic, low-cost security monitoring prototype.

3. COMPONENTS REQUIRED

Component	Purpose
ESP32 development board	Main controller; reads the PIR sensor output and handles the program logic.
PIR motion sensor	Detects changes in infrared radiation associated with movement.
LED	Provides a visual indication in the prototype; its exact control behaviour depends on the uploaded code and wiring.
Breadboard	Allows components to be connected without soldering.
Jumper wires	Provide electrical connections between the ESP32, sensor, and indicator.
USB cable and computer	Power/program the ESP32 and view messages in the Arduino IDE Serial Monitor.

4. WORKING PRINCIPLE

A PIR sensor detects changes in infrared energy in its field of view. When a warm body moves across the sensing area, the sensor detects the change and produces a digital signal. In the usual digital-output configuration, HIGH (1) indicates detected motion and LOW (0) indicates that motion is not currently being detected. The ESP32 reads this signal repeatedly and prints the corresponding status message to the Serial Monitor.

- No motion: PIR output is LOW (0); the Serial Monitor displays “NO MOTION”.
- Motion detected: PIR output is HIGH (1); the Serial Monitor displays “MOTION DETECTED!”.

5. HARDWARE SETUP

The ESP32 board and PIR sensor are connected using jumper wires on a breadboard. The board is powered through the USB cable connected to the computer. An LED is mounted on the breadboard as a visible indicator. The photographs supplied with this report show the assembled prototype. The exact GPIO pin numbers and power connections should be recorded from the final circuit or source code before submitting a pin-by-pin wiring table.

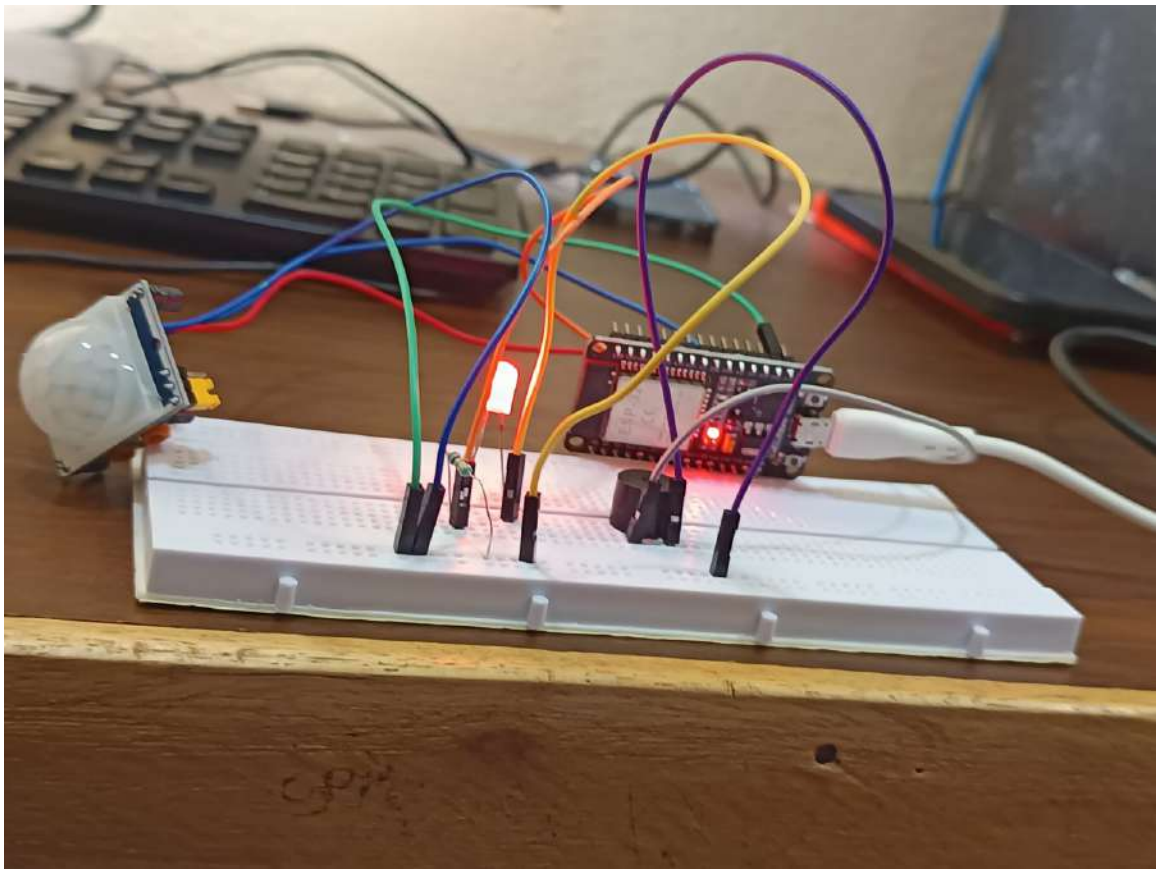


Figure 2: Experimental hardware arrangement

6. SOFTWARE AND IMPLEMENTATION

The ESP32 is programmed using the Arduino IDE. The program configures the PIR-connected GPIO pin as an input and starts serial communication. Inside the main loop, it reads the sensor using a digital input operation. If the reading is HIGH, the program prints the motion-detected message; otherwise, it prints the no-motion message. A short delay may be used between readings to make the output easier to observe. The precise pin definitions, delay value, and LED logic should match the actual source code used in the prototype.

7. TESTING AND OBSERVATIONS

The system was powered through USB and monitored using the Arduino IDE Serial Monitor. The captured output contains both sensor states. A reading of 0 is followed by “NO MOTION”, while a reading of 1 is followed by “MOTION DETECTED!”. This indicates that the ESP32 is receiving changing digital readings from the PIR sensor and displaying the corresponding status.

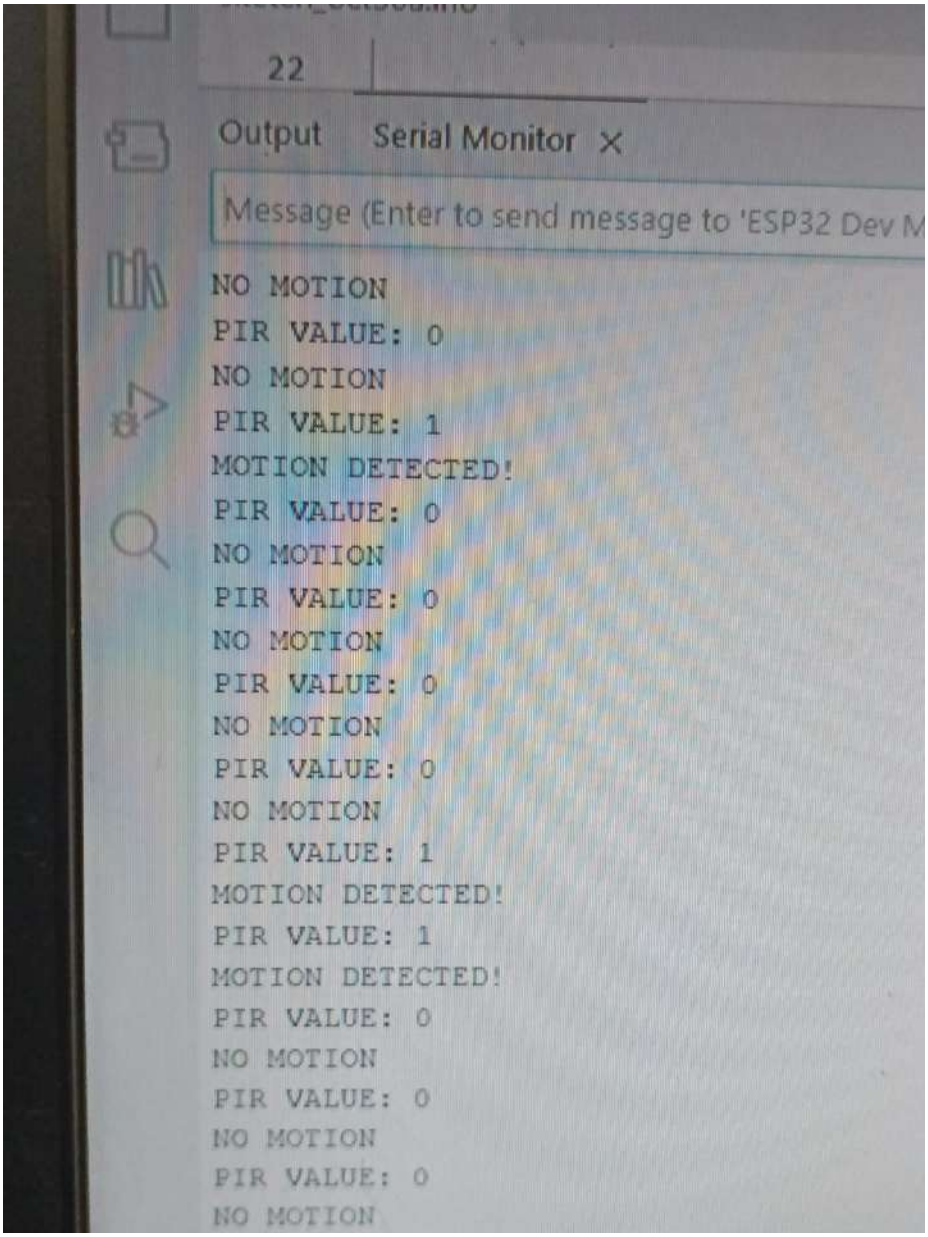


Figure 3: Serial Monitor output showing PIR values 0 and 1

PIR value	Condition	Observed message
0 (LOW)	No motion detected	NO MOTION
1 (HIGH)	Motion detected	MOTION DETECTED!

8. ADVANTAGES

- Simple and easy-to-understand working principle.
- Uses an affordable and widely available motion sensor.
- ESP32 provides a convenient platform for future IoT extensions.
- Serial Monitor output makes testing and debugging straightforward.
- The prototype can be expanded with an alarm, notifications, or network connectivity.

9. APPLICATIONS

- Basic room and office movement monitoring.
- Laboratory and equipment-room monitoring.
- Entry-area or corridor motion detection.
- Educational demonstrations of sensors and microcontrollers.
- A starting point for IoT-based home security systems.

10. LIMITATIONS AND PRECAUTIONS

- A PIR sensor detects changes in infrared radiation; it does not identify a person.
- Detection range and sensitivity depend on the sensor module and its adjustment.
- The sensor may require a short stabilisation period after power-up.
- The system should be tested under different movement directions and environmental conditions.
- Check the ESP32 GPIO voltage limits and the sensor module's output voltage before connecting them.
- This prototype is a demonstration and should not be treated as a complete certified security alarm.

11. FUTURE SCOPE

The prototype can be improved by adding a buzzer or siren for an audible alert, sending notifications over Wi-Fi using the ESP32, logging motion events with timestamps, and adding a mobile or web dashboard. A camera module or additional sensors could also be integrated for more advanced monitoring.

12. CONCLUSION

The Motion Detection Based Security System using ESP32 and a PIR sensor was assembled and tested. The Serial Monitor output shows that the system reports “NO MOTION” when the PIR output is 0 and “MOTION DETECTED!” when the output is 1. The observed results demonstrate the basic operation of motion sensing and digital input processing using an ESP32. This project provides a useful foundation for developing more advanced IoT-enabled security applications.

REFERENCES

1. Espressif Systems, ESP32 Series Datasheet and Technical Documentation.
2. Arduino Documentation, Digital Input/Output and Serial Communication.
3. PIR motion sensor module datasheet or module documentation (refer to the exact module used).

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